Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)			Correct symbols for lamp and cell and variable resistor shown correctly in a series circuit (1) **Accept** variable power supply instead of cell and variable resistor If more than 1 cell shown they must be connected correctly in terms of polarity and no line passing through the cell Correct symbol for voltmeter and connected in parallel with their symbol for a lamp only (1) Correct symbol for ammeter and connected in series (1)	3			3		3
	(b)	(i)		Scale on x-axis of 1 V/cm **and** on y-axis of 0.1 A/cm (1) **Accept** scale on y-axis of 0.125 A/cm All points correctly plotted ± <1 small square division – ignore (0,0) plot (1) Less than 5 points correctly plotted ± <1 small square division – ignore (0,0) plot (0) Smooth curve between 0 – 12 V with ± <1 small square division (1)		3		3	3	3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Any one calculation of resistance e.g. $R = \frac{2}{0.4}$ = 5 [Ω] (1) Another calculation of resistance e.g. $R = \frac{12}{1.1}$ = 10.91 [Ω] (1) so disagree – conclusion must be consistent with results and based on at least **2 calculated values** of resistance (1) For information: Voltage (V) | Resistance (Ω) 2 | 5[.00] 4 | 5.26 6 | 6.38 10 | 9.26 12 | 10.91 Ignore incorrect rounding of resistance values Use of $R = \frac{P}{I^2}$ to obtain correct values award the marks			3	3	3	3
				Question 3 total	**3**	**3**	**3**	**9**	**6**	**9**

For information:

Voltage (V)	Resistance (Ω)
2	5[.00]
4	5.26
6	6.38
10	9.26
12	10.91